

Sudden transition to rapid warming in future projections of the Peruvian coastal upwelling system

The Projekt Woche Team¹

¹Max Planck Institute for Meteorology, Hamburg, Germany

Key Points:

- The evolution of the Peruvian coastal upwelling system is analyzed in global, kilometer-scale coupled climate simulation
- Despite global warming, upper ocean temperatures remain stable as long as the simulation sustains a cooling trend in the Southern Ocean and Eastern Pacific, as seen in observations
- A sudden shift occurs after 2020, when the global pattern of warming changes and enhanced ocean eddies suppress the ability of upwelling to cool the upper ocean

Corresponding author: =name=, =email address=

Abstract

Southerly winds that blow along the Peruvian coast induce ocean upwelling that helps cool the local sea surface temperature (SST). Observations and climate models offer contrasting accounts of how the Peruvian coastal upwelling system (PCUS) might respond to global warming. Historical data show a steady trend towards faster winds and colder SST, both consistent with an strengthening PCUS. In contrast, simulations typically project rapid warming under radiative forcing by greenhouse gases. Qualitative disagreement between observations and models therefore sows uncertainty about the future evolution of climate in this region and its possible implications to biological productivity and CO₂ outgassing. **Here, we present output from a kilometer-scale global coupled climate simulation featuring both regimes, with a decades-long trend of SST cooling followed by rapid warming.** The transition occurs around the simulation's year 2020, when the Peruvian coast warms up even though the upwelling circulation strength is at near-peak values. Timing of this transition coincides with a large-scale shift that leads to warming of the surrounding waters that feed the PCUS and thereby suppress the cooling effects of advection. Our results suggest that, despite historical disagreement between simulated and observed coastal SSTs, the Peruvian coast may indeed transition to a state of rapid warming.

Plain Language Summary

Historical data show that waters off the Peruvian coast have cooled steadily, despite widespread global warming. However, simulations of this region's climate suggest rapid local warming under radiative forcing by greenhouse gases. A new global climate simulation run at unprecedented spatial resolution reveals both trends in sequence: a long period of coastal cooling followed by a sudden shift to rapid warming around the year 2020. *Simulated warming occurred even while coastal upwelling remained strong, as a large-scale climate shift accelerated the warming of ocean currents that flow towards the Peruvian coast.*

1 Introduction

Southerly winds that blow along the Peruvian coast induce off-shore mass transport at the ocean surface that leads to upwelling of cold, nutrient-rich subsurface waters. This circulation helps cool the Peruvian coast and contributes to the outstanding primary productivity of its waters. Interannual variations in along-shore winds, SST, and upper ocean stratification exemplify how regional weather and human well-being are particularly sensitive to the state of the Peruvian coastal upwelling system (PCUS). Therefore, understanding the likely PCUS response to global warming is crucial to the development of adaptation and mitigation strategies across fisheries, health, and civil safety.

A majority of coupled climate simulations suggest that large scale trends in surface winds, driven by strengthened subtropical highs and enhanced land-sea contrasts, will lead to intensified coastal upwelling globally under global warming (Wang et al., 2015). However, the sign of trends in upwelling-favorable wind stress over the PCUS varies across datasets (Narayan et al., 2010). Even in a future scenario of strengthened winds and upwelling, constraining the likely evolution of SST and upper ocean stratification requires general circulation models with horizontal resolution ~ 10 km (Capet et al., 2004; Colas et al., 2012). This is far higher than the standard resolution for global coupled climate simulations, which typically run on grids with horizontal spacings in the order of 50 to 100 km.

Until now, only regional uncoupled simulations have enabled detailed projections of the PCUS response to global warming. Such model projections are usually forced with downscaled atmospheric products and consistently indicate coast-intensified SST warm-

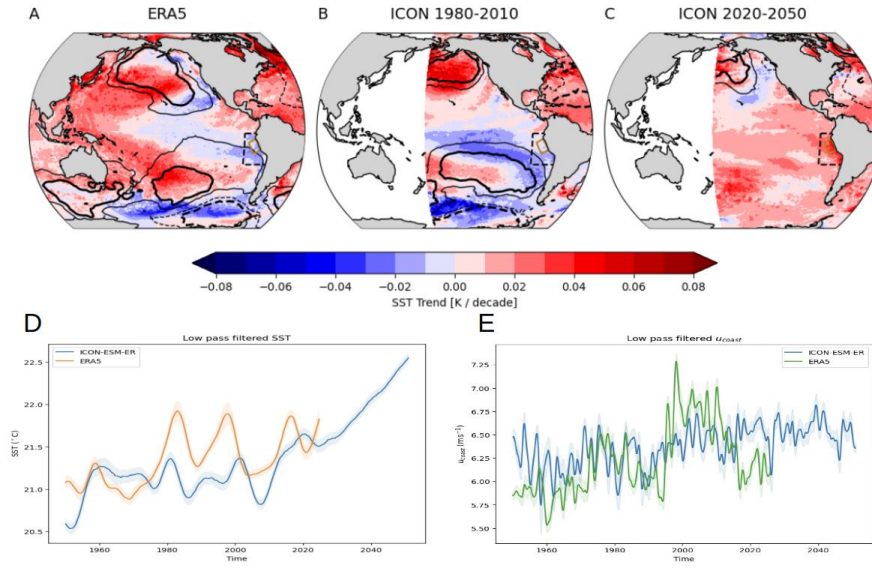


Figure 1. Large scale trends in the Pacific Ocean and Peruvian coast. Linear trends in (color shading) SST and (black contours) sea level pressure are computed from A) the ERA5 reanalysis and B-C) the **ICON simulation**. Panels A) and B) show trends computed between 1980 and 2010, while C) uses simulation output between 2020 and 2050. Timeseries show the long-term evolution of D) SST and E) along-shore winds in the PCUS region. Both records are lowpass-filtered using a 10-year Hanning window.

ing (Oerder et al., 2015; Echevin et al., 2020). Most of these simulations feature a weakening of wind-driven upwelling, and thus associate coastal SST warming to a weakening of the PCUS circulation. Unfortunately, the consistency of PCUS projections across regional models clashes with understanding from theory, global coupled models, and observational trends alike (Bakun, 1990; Narayan et al., 2010; Gutiérrez, Bouloubassi, et al., 2011; Wang et al., 2015; Briceño-Zuluaga et al., 2023). All these issues complicate interpretation of model results and hinder our ability to understand the changing dynamics of coastal upwelling systems off Perú and elsewhere.

Observations show that SST have persistently cooled since the mid-20th century over most of the PCUS (Gutiérrez, Bouloubassi, et al., 2011; Gutiérrez, Bertrand, et al., 2011). Upwelling-favorable winds have strengthened over an even longer period (Vargas et al., 2007; Briceño-Zuluaga et al., 2023) and would thus suggest that a strengthened PCUS has suppressed any possible warming. By now, greenhouse gases have clearly reshaped countless aspects of the global climate system, but the PCUS continues to evolve in a direction contrary to model projections. This raises the question: are model projections of coastal SST warming wrong, or has the historical record been dominated by strong internal variability that masks a weaker warming trend?

Model errors may arise from multiple sources, but we highlight two. First, from the fact that regional downscaling experiments of the PCUS under global warming inherit the large scale biases of the global models used to force their boundary conditions. This is particularly concerning because a majority of global climate models fail to reproduce, or underestimate, a large scale historical trend of cooling across the Southeast Pacific Ocean (Watanabe et al., 2024). As a result, downscaling simulations may embed the PCUS within unrealistically high regional temperatures and bias results towards coastal SST warming under greenhouse gas forcing. Second, downscaling simulations are

forced using predetermined daily atmospheric fields rather than coupled to the evolving atmosphere (Oerder et al., 2015; Echevin et al., 2020). This lack of coupling may lead to a biased representation of ocean eddies that play a significant role in regulating Peruvian coastal SSTs and upwelling velocities (Colas et al., 2012; Oerder et al., 2018, 2024).

Here, we overcome some of the main obstacles that have hindered previous studies of the PCUS response to global warming. We present a historical simulation and future projection of the PCUS and its SST from a global kilometer-scale coupled climate simulation between 1950 and 2050. Model runs have horizontal resolutions of 5 km in the ocean and 10 km in the atmosphere (Hohenegger et al., 2023), sufficient to resolve ocean eddies and salient aspects of their interaction with nearshore meteorology. The simulation is particularly unique because it reproduces historical large scale trends over the Pacific Ocean with unprecedented skill (Kang et al., 2026).

Model results reveal a decades-long trend of slight cooling that is consistent with historical observations. However, this trend is followed by a sudden transition to rapid coastal-intensified warming as projected by previous regional models. Upwelling-favorable winds and the PCUS circulation both strengthen throughout the simulation, but the cooling effect of ocean advection eventually subsides to warming of the broader environment that surrounds the PCUS. This emphasizes that both the cooling trend in observations and the warming response in previous simulations can coexist within a single simulation. Therefore, historical strengthening of the PCUS may continue onto the future but will not necessarily always shield the Peruvian coast from accelerated warming that has already taken over vast regions of the global ocean.

The manuscript is structured as follows.

2 Methods and data

Fraser is writing a model description section. It will talk about ICON, its grid, and mention that it is part of the EERIE project.

We may also want to include a quick description of the advective heating estimate. Maybe Andrea can write that?

We might as well add one single paragraph explaining what we designate as an "eddy" (sub-monthly motions) and mention that is not necessarily the definition that others have used but we will stick with it.

3 Results

3.1 Rapid warming under strong upwelling conditions

Simulated values of SST over the PCUS remained mostly stable or cooled slightly between 1960 and 2010, but then warmed rapidly (blue line in Fig. 1D). This sudden change in evolution coincides with a broader, large scale shift in SST trends across the Pacific Ocean (Figs. 1B,C). Between 1980 and 2010, SST cooled across the Southeastern Pacific and Southern Oceans, while sea level pressure strengthened off the South American coast. This trend helped accelerate along-shore winds (u_{shore}) in the PCUS, which strengthened by roughly 10% throughout the simulation (Fig. 1E). SST trends reversed sign after 2020, when warming was strongest near the Peruvian coast even though SLP changes were not undone and u_{shore} remained at all-time-highs (Figs. 1C,E).

A close-up view of simulated SST trends between 1980 and 2010 reveal a meridional gradient in warming, with the southern half of the PCUS cooling, and the northern half warming (Fig. 2A). This meridional contrast matches previous reports based on in-situ measurements (Gutiérrez, Bertrand, et al., 2011). Furthermore, the overall mag-

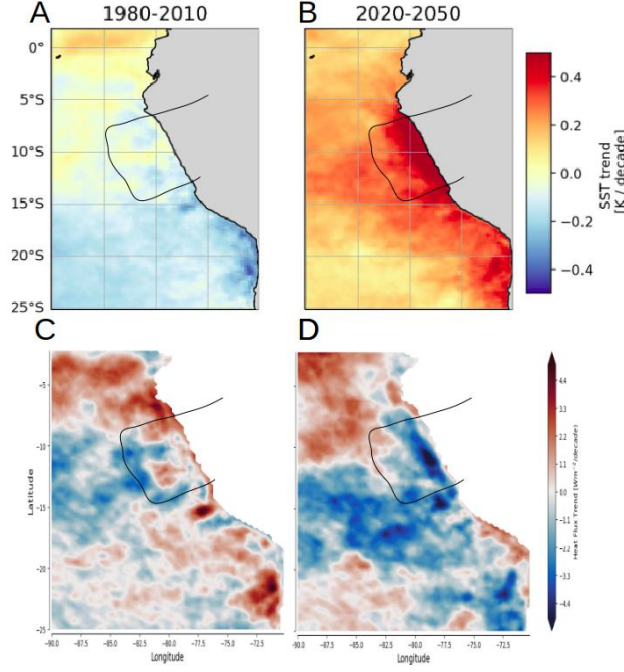


Figure 2. Linear trends in simulated A-B) SST and C-D) Q_{net} . Trends are computed for the periods between A,C) 1980-2010 and B,D) 2020-2050. The black rectangle outlines our designation of the PCUS. *Here I want to show the change in mean heat flux, rather than its trend.*

well?
 nitude and evolution of SST and u_{shore} compares were with reanalysis values (Figs. 1D,E). Such favorable comparisons between the ICON simulation and the historical record demonstrate the model's skill and help inform the interpretation of its future projections.

It is surprising that southeast Pacific SST warming trends would peak along the Peruvian coast when u_{shore} sustained maximum strength (Fig. 2B). This counterintuitive pattern of warming may not be explained by atmospheric heat fluxes (Q_{net} , positive into the ocean), which weakened between the historical and future periods than before (Fig. 2C). Therefore, only subsurface ocean processes may explain the simulated transition to sudden PCUS warming.

Cross-shore sections of subsurface ocean temperature (T) and currents (u, w) in Fig. () show the simulated circulation of the PCUS and its changes between the historical and future periods. As the changes in u_{shore} suggested, offshore transport and coastal upwelling intensified over time, but the near-coastal T became warmer. *(I wrote this based on the trends in T and UT in Fig. X. However, I would like it better if we showed the average T [color shading] and u,w [arrows] as well as their changes rather than the trends).*

Let's add a row to Fig. 3 showing the along-shore velocity. This way we can also say something about local changes in the Humboldt current system. Maybe the undercurrent that comes from warmer, higher latitudes is becoming shallower and moving closer to the shelf? The point is to get a full description of how the local circulation is changing.

Here is where we present timeseries of the divergence of heat transport within the PCUS region. Ideally we can separate them between eddies and mean circulation.

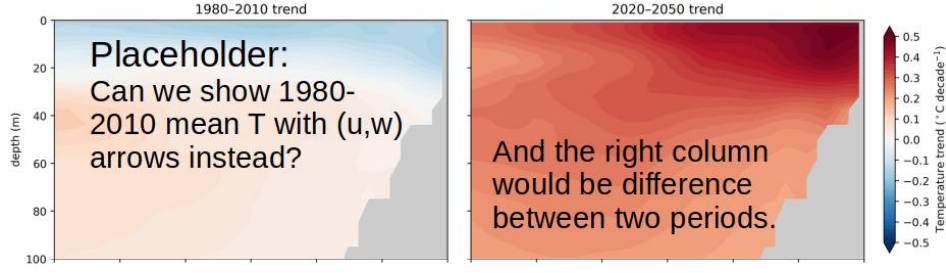


Figure 3. This is a placeholder. The text shows my proposal for a figure showing the 1980-2010 mean temperature, stratification, and circulation, as well as their changes for 2020-2050 (right column). A second row could show color shading for stratification and contours for along-shore velocity. Or maybe we can use the second row to jump straight to the time series of heat transport divergence. Sections are averaged within the black rectangle highlighted in Fig. 2.

We might want to add a subsection that further separates heating terms and changes into seasons. I just noticed that almost all of the decrease in Q_{net} comes from changes in May-July, when u_{shore} and upwelling are the strongest. This might make upwelling particularly inefficient to cool the ocean surface at that time, as per the rationale of Thomsen.

3.2 Long-term changes are concentrated in the upwelling season

Strong seasonality of air-sea interactions in the PCUS was well-represented in the simulation, with u_{shore} peaking between May and September when Q_{net} cools the ocean surface (Fig. 4). Long-term changes in atmospheric forcing also have their own seasonality, as the increased u_{shore} evidenced in Fig. 1E is entirely explained by changes between February and July (Fig. 4A). Similarly, the decrease in Q_{net} over the PCUS is primarily associated with enhanced cooling of the ocean surface between May and August (Fig. 4B). Therefore, broad changes to the atmospheric forcing of the PCUS were greatest during the Winter season when the upwelling circulation is strongest.

Once again, intuition would suggest that concurrent strengthening of u_{shore} and weakening of Q_{net} during the upwelling-favorable season (May-September) would yield a colder PCUS. However, local SST warming relative to 1980-2010 was maximum in May (Fig. 4C). This emphasizes that upwelling's cooling influence on SST is highly dynamic and may be suppressed in a future scenario.

3.3 The changing role of eddies

Baroclinic fronts separate cold surface waters in the PCUS from warmer regions farther north and off-shore from the coast of Peru. Eddies can extract energy from sloping isopycnals along these fronts and induce lateral transports that restratify the upper ocean (Brink, 2016). This way, eddies can suppress the ability of upwelling the cool SST (Thomsen et al., 2021), but also disperse nutrients away from the coast and weaken local biological productivity (Gruber et al., 2011).

Eddy kinetic energy (EKE) in our simulations strengthened by $\sim 3 \text{ J m}^{-3}$ off the Peruvian coast, up $\sim 40\%$ from values $\sim 7 \text{ J m}^{-3}$ between 1980 and 2010 (Figs. 5A,B). Onshore heat fluxes by eddies opposed the heat transport direction by the mean circulation and increased over the same period (Figs. 5C,D). Along with greater values $\langle w'T' \rangle$, this indicates that warm sub-monthly anomalies were consistently advected upwards and towards the coast, thus helping restratify the shelf.

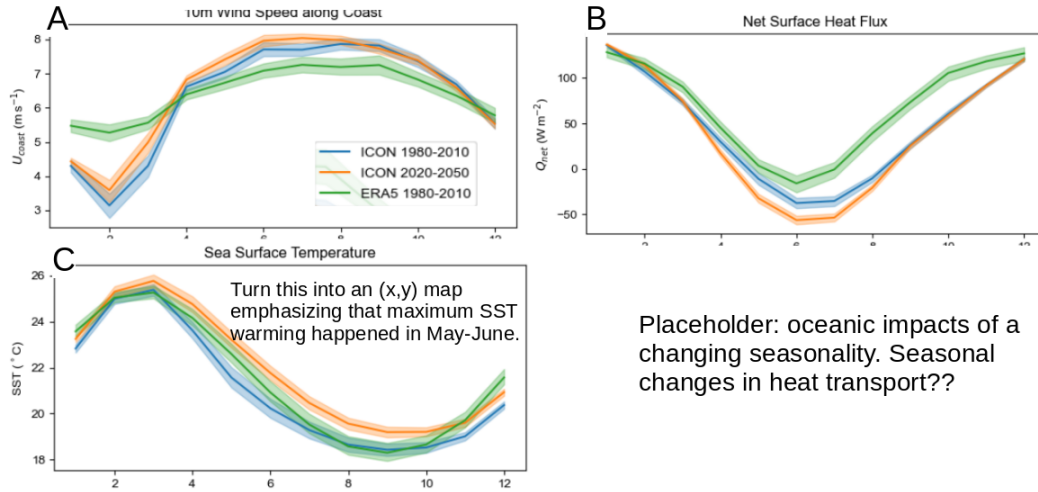


Figure 4. Seasonality of atmospheric forcing and heat transport in the PCUS. Time series show the seasonal cycles of A) along-shore winds u_{shore} and B) air-sea heat flux Q_{net} , both averaged for the ocean region outlined in Fig. 2. Values are shown separately for the (green) ERA5 reanalysis, (blue) ICON simulation between 1980 and 2010, and (orange) ICON between 2020 and 2050. Maps in C) and D) may show vertically-averaged changes in net ocean heat transport and SST changes for the upwelling season (May-August).

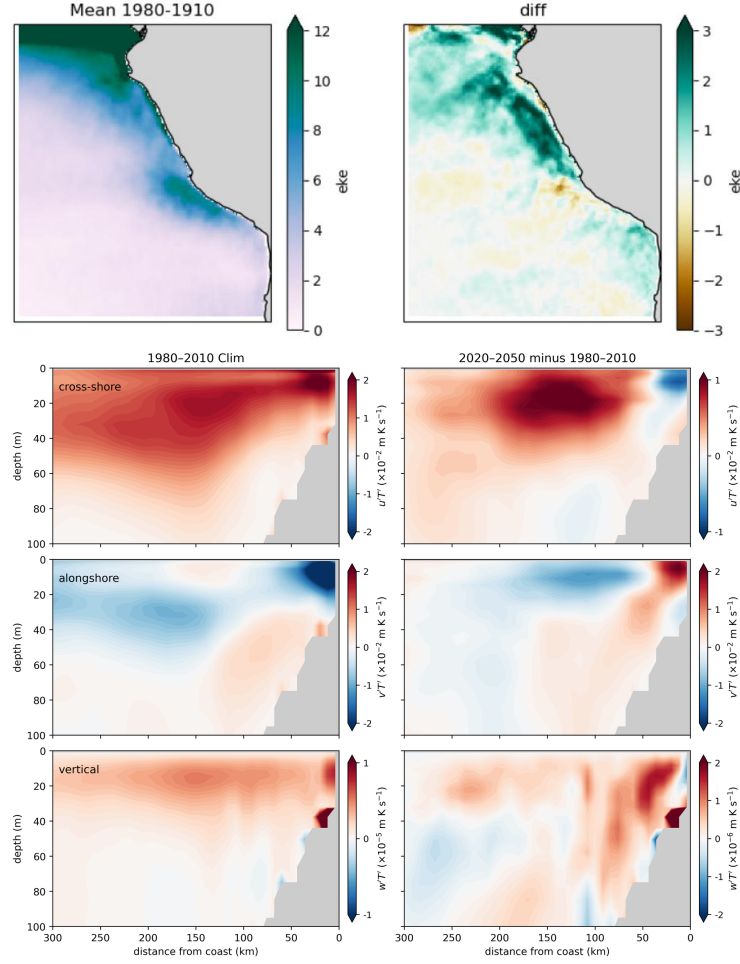
Meridional heat fluxes by eddies helped transport warm, near-equatorial waters towards the PCUS but this influence moved offshore throughout the simulation (color shading in Figs. 5C,D). This suggests that the restratification effects of eddies acted across east-west and north-south SST fronts that engulf the Peruvian coast. Decadal averages of SST (Figs. 5A,B) show that the XX and YY isotherms moved by roughly ZZ km south and towards the Peruvian coast,

3.4 Air-sea interaction effects of eddies

1. EKE and $Q'T'$ shown side to side. More eddy activity when eddy damping by $Q'T'$ weakens.
2. We are not claiming that $Q'T'$ are the only process driving the multidecadal evolution in EKE. However, we note that simulating the evolution of

4 Conclusion

1. We've shown a regime shift. It suggests that, as long as the historical large scale pattern continues, the Peruvian coastal system may remain cool. If Southern Ocean cooling stops, our simulations suggest that a sudden transition to warming may occur.
2. Enhanced stratification that comes by with this transition would prevent vertical transport of nutrients and severely impact the biological productivity here.
3. Simulations of the future of upwelling regions require representation of eddies



change to $u'T'$, $w'T'$ as arrow
 $v'T'$ as shading, combining into
 just 3 figures

Figure 5. Mean values (left column) and changes in eddy activity (right column). A) shows mean values of EKE in the surrounding region, while B) shows changes as EKE future - historical. Cross-shore sections in C) and D) show three-dimensional heat transport by eddies in the PCUS and its changes over the same period. Let's try and condense the transport figures into two panels, one for mean and one for change. Color shading can show $v'T$, and arrows can show $u'T$. We will have to scale $w'T$ by some number and include arrows for scale. Let's also overlay SST contours on panels A,B.

Open Research Section

This section MUST contain a statement that describes where the data supporting the conclusions can be obtained. Data cannot be listed as "Available from authors" or stored solely in supporting information. Citations to archived data should be included in your reference list. Wiley will publish it as a separate section on the paper's page. Examples and complete information are here: [https://www.agu.org/Publish with AGU/Publish/Author Resources/Data for Authors](https://www.agu.org/Publish-with-AGU/Publish/Author-Resources/Data-for-Authors)

Conflict of Interest disclosure

The authors declare there are no conflicts of interest for this manuscript.

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Enter acknowledgments here. This section is to acknowledge funding, thank colleagues, enter any secondary affiliations, and so on.

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